THE PRESIDENT'S OFFICE

REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT

LONAE PRE-MOCK JOINT EXAMINATION

FORM FOUR

BASIC MATHEMATICS

CODE; 041
TIME; 3 HRS **04TH MAY 2021 08:00Am – 11:00 Am**
INSTRUCTIONS

1. This paper consists of section A and B with total of 14 questions
2. Answer all questions in both sections *A* and *B*
3. Mathematical table may be used if necessary

SECTION A: (60 Marks)

Answer all questions in this section

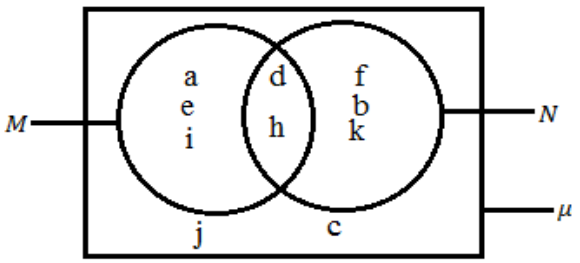
1. (a) If $a = 0.\dot{2}\dot{5}$ and $b = 0.\dot{5}$, find the fraction $\frac{a}{b}$ in its simplest form.

   (b) Round off:
   (i) 0.002098 to two significant figures
   (ii) 7.67 to the nearest whole number
   (iii) 2.77 to one decimal place

2. (a) Rationalize the denominator $\frac{2}{2\sqrt{3}+\sqrt{2}}$ in the form of $a\sqrt{x} + b\sqrt{y}$

   (b) Solve for x if $\frac{2^{2x+7}}{4^{2x-2}} = \frac{8^{x-3}}{32^{8-x}}$

3. (a) given the figure below;



   Find (i) $M' \cap N$ (ii) $n(M' \cup N')$

   (b) A box contains 4 white balls and 5 black balls. Two balls are selected at random without replacement. Find the probability that

   (i) Both are white balls

(ii) The first is black and the second is the white ball

4. (a) $\underline{a} = 3\underline{i} + 4\underline{j}$, and $\underline{b} = -2\underline{i} + 6\underline{j}$

   i. Solve for V=$2\underline{a} + 3\underline{b}$

   ii. Find magnitude of V

   (b) (i) What value of K will make the line containing the points (K, 5) and (-2, 3) parallel to the line containing points (6, 2) and (K, 0)?

   (ii) Calculate the distance between points (1,5) and (9, 5).

5. (a) Twice the width of a rectangle is greater than its length by $3cm$, if the perimeter of the rectangle is $36cm$. Find its dimensions

   (b) Find the number of sides of a regular polygon whose sum of its interior angle is 1080

6. (a) If y varies directly to $x - 1$ and inversely to $x + d$ and if $x = 2, d = 4$ and for $y = 1$. Find x when $y = 2$ and $d = 1$.

   (b) How many grams are there in 0.00912 tones.

7. (a) Given that $L: M = 2: 1$ and $M: N = 3: 4$. Find the value of $N: L$

   (b) Define the following; (i) Double entry (ii) Transaction (iii) Assets

8. (a) The fifth term of an AP is 40 and the seventh term of the same AP is 20. Find

   i. Common difference

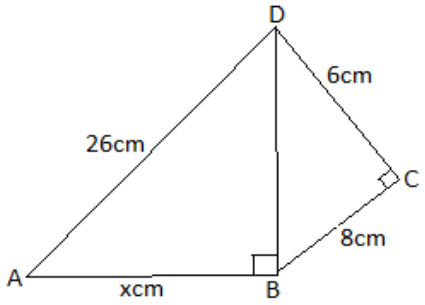
   ii. The 10th term

   (b) Laizer invested a certain amount of money in a Savings Bank whose interest rate was 10% compounded annually. After two years he got 5000 shillings.

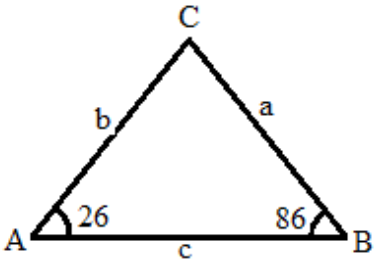
   i. How much did he invest at the start?

   ii. How much did he receives as Interest at the end of two years.

9. (a) Find the length of AB in figure below



   (b) Find unknown sides and angles in ΔABC in which $a = 22.3cm, \hat{B} = 86°$ and $\hat{A} = 26°$.



10. (a) Find the value of ab in $\begin{cases} a^2 + b^2 = 60 \\ (a-b)^2 = 48 \end{cases}$

(b) Find the consecutive even numbers whose product is 80.

SECTION B: (40 Marks)

Answer all questions in this section

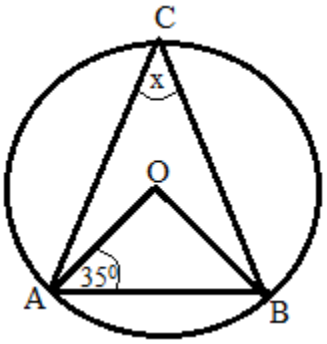
11. (a) The table below shows the height of the students

Height in (cm)	$11-15$	$16-20$	$21-25$	$26-30$	$31-35$	$36-40$
Number of students	4	8	20	21	12	13

(i) Calculate mode

(ii) Draw the cumulative frequency curve and use it to estimate median

(b) Find the angle x in the figure below



12. (a)A ship sails from point A $(40°S, 30°W)$ due west along the same latitude to point B for 1000km. Find the latitude and longitude of point B. Use R=6370km and $\pi = 3.14$ (give your answer in nearest degree)

(b) VABCD is a pyramid with VA=VB=VC=VD=5cm and ABCD is a square base of sides 4cm each. Assume that the centre of the base is at point N. Find

i. The angle between VA and the base ABCD
ii. The volume of the pyramid.

13. (a) Given that $A = \begin{pmatrix} 2 & -3 \\ -1 & 2 \end{pmatrix}$ and $B = \begin{pmatrix} -2 & 6 \\ 1 & 2 \end{pmatrix}$.

i. Find value of 2A + B -5I where I is the identity matrix.

ii. Determine the determinant of matrix in (a) (i) above

(b) Find x and y by inverse matrix method, given the equations;

$x + 3y = 5$ and $2x + 8y = 12$

(c) Find the image of a point $(4, 3)$ when undergoes reflection at the line $x = y$ then translated at a factor $T = (-2, 2)$

14. (a) A function $f(x) = \begin{cases} x \ \ if -3 \leq x < 3 \\ x^2 \ if -3 \leq x < 5 \\ 5 \ if \ x \geq 5 \end{cases}$

(i) Draw the graph of $f(x)$

(ii) State the domain and range

(iii) Find $f(10)$

(b) Suppose that we are given the information in the following table regarding the vitamin content and cost of two kinds of pills as well as the minimum unit of each vitamin required per day

	White pill	Red pill	minimum unit required per day
Vitamin A	4 units	1unit	8
Vitamin B	1 units	1unit	5
Vitamin C	2 units	7unit	20
Cost per pill	Tsh 70/=	Tsh.90/=	

How many of each kind of pills should be taken in order to fulfill minimum daily requirements at the lowest cost?